

Qatar-2b

R-1703

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_190323 · created 2026-08-02T14:26:16+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458565.92216 +/- 0.00077 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.064
Transit depth (Rp/Rs)^2	3.6e-05 +/- 0.0077 [%]
Orbital Inclination (inc)	88.9 +/- 1.71
Airmass coefficient 1 (a1)	0.037 +/- 0.018
Airmass coefficient 2 (a2)	2.49 +/- 1.61
Scatter in the residuals of the lightcurve fit is	43.08 %
Best Comparison Star	#2 - [519, 461]
Optimal Aperture	3.09
Optimal Annulus	5.41
Transit Duration (day)	0.043 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.064 vs 0.16327 (deviation 2.54σ)
Duration fit vs published	0.043 d vs 0.0767 d (deviation 16.83σ)
Mid-time O-C	-1.9 min
Depth SNR	0.0
Residual scatter	43.08 %

Light curve

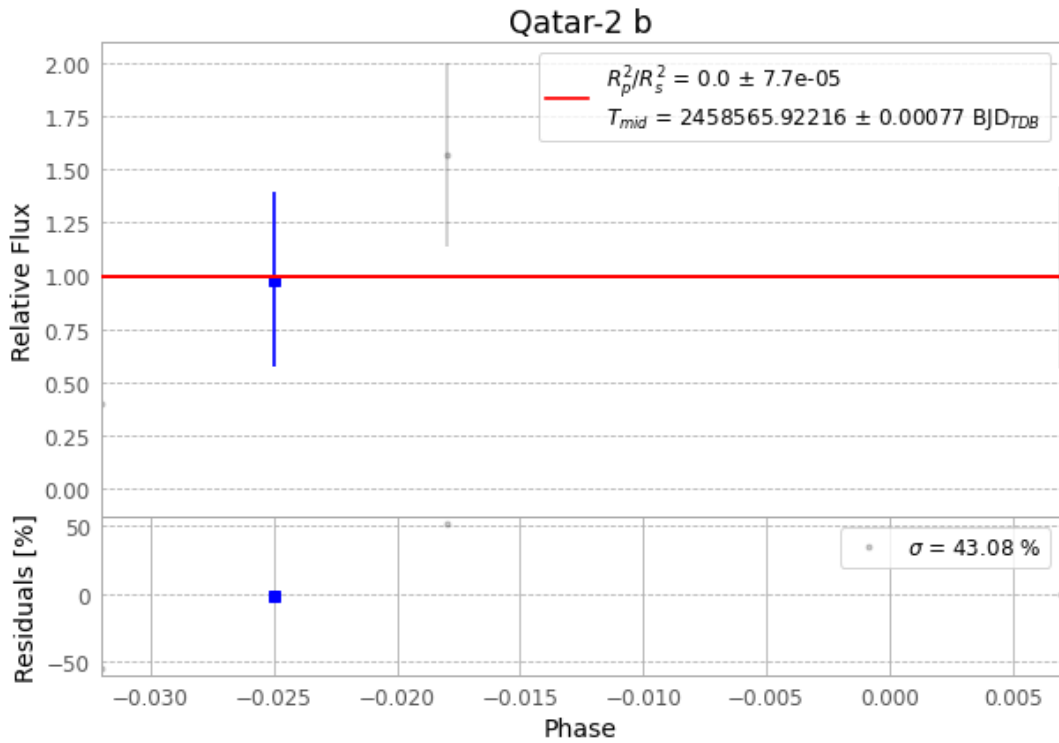


Figure 1 — FinalLightCurve_Qatar-2 b_2019-03-23.png (SHA-256 5d5f8e0c75bcf874...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [532, 352], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d3f6caaa0c52d2da...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

68 FITS frames (45 MB), Qatar-2190323082412.FITS → Qatar-2190323114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-190323.log (2071e9179fa2...), FinalLightCurve_Qatar-2 b_2019-03-23.png (5d5f8e0c75bc...), FinalLightCurve_Qatar-2 b_2019-03-23.csv (86736728e18e...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 14:26Z.